Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/13/2006
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: End date changed from 12/31/2006 to 9/30/2006 (1/2009) NOTE: grant to be reinstated with a new grant number. 8/2006, per D. Urban		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1063		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. Recognizing the unique benefits of spherical diffusion flames, employ such flames to: 1. Obtain soot-inception limits as functions of stoichiometric mixture fraction, flow direction (direction across the flame front, i.e., towards the fuel or towards the oxidizer), amount of inert (nitrogen) and pressure for ethylene and methane flames. Identify the corresponding soot-inception temperature; 2. Determine effects of stoichiometric mixture fraction and flow direction on the phenomenon of permanently blue flames and, to this end, determine the effects of stoichiometric mixture fraction and flow direction on flame temperature, size, color and soot volume fraction; 3. Obtain extinction limits as functions of stoichiometric mixture fraction, flow direction, amount of inert and pressure for ethylene and methane flames. Identify the corresponding extinction temperature; 4. Evaluate whether this flame system can be used to obtain a pseudo-flammability limit for non-premixed flames in terms of stoichiometric mixture fraction and amount of inert. If so, determine these limits for ethylene and methane flames. Soot formation and flame extinction are problems of great concern in practical combustors. Recent normal-gravity experiments using counterflow diffusion flames have shown that permanently-blue flames (which are devoid of soot regardless of strain rate) and extinction-resistant flames can be produced for many hydrocarbon fuels by nitrogen exchange (increased stoichiometric mixture fraction). Furthermore, there is a unique stoichiometric mixture fraction wherein the flame temperature at extinction can be up to 200 K lower and the strain rate up to four times greater than that for the standard hydrocarbon/air flame with the same flame temperature. The present experiments employ microgravity conditions to allow for a unique set of experiments to clarify the underlying mechanisms responsible for flame response to stoichiometric mixture fraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As the goal of this program is to develop an understanding of combustion with the aim of reducing pollutants and increasing combustion efficiency, the results are directly applicable to terrestial power generation. In particular, with the present interest in carbon sequestration via oxygen-enriched combustion the present work is of particular importance as it identifies methods of controlling a number of pollutants through oxygen-enriched combustion.
Task Progress:	Below is a summary of the progress on this project for the year ending January 1, 2006: Flame Design benefited from several analytical and computational advances in 2005. These have yielded important insight into many experimental observations by this project. This has laid the groundwork for two journal publications, one which was submitted and one which is nearing submission. Work on the general behavior of spherical diffusion flames has demonstrated that a detailed transient numerical model (with detailed chemistry, transport, and radiation) can predict flame size and flame temperature for microgravity spherical diffusion flames. A simplified model indicates that Lewis number can have a large effect on flame temperature for microgravity diffusion flames. This work has been submitted to the Combustion Symposium. Work on radiative extinction of spherical diffusion flames has demonstrated that the detailed transient model can predict extinction times for microgravity spherical diffusion flames. The work identified several scaling relationship for extinguishing diffusion flames. This work is being finalized for an upcoming journal submission. A Nikon D100 color digital still camera has been configured for use in the 2.2 second drop tower. This provides four images per drop. The images have 6 megapixels and 12 x 3 bit depth and are among the highest-fidelity images ever recorded in the drop tower. This camera has been used to record images of 15 micron SiC fibers in flames in the 2.2 second drop tower. This has yielded qualitative thin-filament pyrometry results for extinguishing spherical diffusion flames. Eighteen successful drop tests were performed in the 2.2 second drop tower.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Sunderland PB, Urban DL, Stocker DP, Chao BH, Axelbaum RL. "Sooting limits of microgravity spherical diffusion flames in oxygen-enriched air and diluted fuel." Combustion Science and Technology. 2004 Dec;176(12):2143-2164. http://dx.doi.org/10.1080/00102200490514994 , Dec-2004
Articles in Peer-reviewed Journals	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." Combustion Theory and Modelling. 2007 Aug;11(4):639-52. http://dx.doi.org/10.1080/13647830601161567 , Aug-2007
Articles in Peer-reviewed Journals	Liu S, Chao BH, Axelbaum RL. "A theoretical study on soot inception in spherical burner-stabilized diffusion flames." Combustion and Flame 2005 Jan;140(1-2):1-23. http://dx.doi.org/10.1016/j.combustflame.2004.08.016 , Jan-2005
Papers from Meeting Proceedings	Santa KJ, Chao BH, Sunderland P, Taylor JL, Urban DL, Stocker DP, Axelbaum RL. "Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, Jan. 9-12, 2006 . AIAA Paper AIAA-2006-747, 9 p. January 2006. , Jan-2006

Papers from Meeting Proceedings	Sunderland PB, Taylor JL, Santa KJ, Chao B-H, Urban DL, Stocker DP, Axelbaum RL. "Extinction of Normal and Inverse Spherical Diffusion Flames in Microgravity." Presented at 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 6 p. Abstract: http://ci.confex.com/ci/2005/techprogram/P1370.HTM , Mar-2005
Papers from Meeting Proceedings	Sun Z, Axelbaum RK, Chao B-H, Santa KJ, Sunderland PB, Urban DL, Stocker DP. "Numerical and Experimental Observations of Spherical Diffusion Flames." Presented at 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. 6 p. Abstract: http://ci.confex.com/ci/2005/techprogram/P1900.HTM , Mar-2005